

7-Day TypeScript Mastery Plan

Day 1 – Core Basics

- What TypeScript is and why it matters
- Setup (tsc, tsconfig.json, integration with Node/React)
- Basic types: string, number, boolean, any, unknown, never, void
- Type inference & type annotations
- Practice: Write a function with typed arguments and return type

Day 2 – Functions & Objects

- Function return types, optional & default params
- Object typing, readonly, optional properties
- Arrays, Tuples
- Enums vs Literal types
- Practice: Create a user profile type with optional fields

Day 3 – Interfaces & Type Aliases

- interface vs type
- Extending interfaces, intersection types (&)
- Union types (|)
- Narrowing with typeof, in, instanceof
- Practice: Define Shape interface and calculate area function

Day 4 – Advanced Types

- Generics ()
- Utility types: Partial, Pick, Omit, Record, Readonly, ReturnType
- Mapped types
- Conditional types (T extends U ? X : Y)
- Practice: Create generic ApiResponse

Day 5 – Classes & OOP

- Classes, public/private/protected, readonly
- Implements (class implements interface)
- Abstract classes
- Getters/Setters
- Practice: Model User and Admin classes

Day 6 – TypeScript with Real Projects

- TypeScript + React: Props & State types

- TypeScript + Node/Express
- Type assertions & guards
- Module systems (ESM, CommonJS)
- Practice: Build small CRUD API or React component

Day 7 – Deep Dive & Patterns

- Discriminated Unions, Exhaustive checks with never
- Declaration merging
- Ambient declarations (.d.ts)
- Config best practices (strict, noImplicitAny)
- Practice: Refactor a JS project fully into TypeScript